

17 Variational autoencoders

$$17.1 \quad \lambda_1, \dots, \lambda_5 \Rightarrow 5 + 10 = 15$$

$$\text{and } \mu_1, \sigma_1^2, \dots, \mu_5, \sigma_5^2$$

$$17.2 \quad f \text{ concave} \Leftrightarrow \frac{\partial^2 f}{\partial x^2} \leq 0 \quad \forall x$$

$$g(x) = \log(x) \Rightarrow \frac{\partial g}{\partial x} = \frac{1}{x}$$

$$\Rightarrow \frac{\partial^2 g}{\partial x^2} = -\frac{1}{x^2} \quad \text{and } -\frac{1}{x^2} < 0 \quad \forall x \in \mathbb{R} \setminus \{0\}$$

$$-\frac{1}{x^2} = 0 \quad \text{for } x=0 \quad \square$$

$$17.3 \quad g(\mathbb{E}[y]) \leq \mathbb{E}[g(y)] \quad \text{if } g \text{ convex}$$

$$g \text{ convex} \Leftrightarrow \frac{\partial^2 g}{\partial x^2} \geq 0 \quad \forall x$$

$$g(x) = x^{2n} \Rightarrow \frac{\partial g}{\partial x} = 2n x^{2n-1}$$

$$\Rightarrow \frac{\partial^2 g}{\partial x^2} = 2n(2n-1)x^{2n-2}$$

$$= 4n^2 x^{2n-2} - 2n x^{2n-2} = (4n^2 - 2n) x^{2n-2}$$

$$\text{proof by induction: } n=1 \quad (4-2)x^0 = 2 \geq 0$$

$$n \rightarrow n+1: (4(n+1)^2 - 2(n+1))x^{2(n+1)-2}$$

$$= (4n^2 + 4n + 4 - 2n - 2)x^{2n}$$

$$= (4n^2 + 2n + 2)x^{2n} \geq 0$$

As x^{2n} positive $\forall n \in \{1, 2, \dots\}$ (even number),
this holds!

$$\text{hence } \mathbb{E}[x^2] \leq \mathbb{E}[x^4].$$

Set $n=1$. Then $g: x \rightarrow x^2$ and the above follows immediately $\square$

$$17.4 \quad \mathbb{E}[\log(\bar{p}(\bar{z}|\theta))] = \int q(\bar{z}|\theta) \log\left[\frac{\bar{p}(\bar{z}|\theta)}{q(\bar{z}|\theta)}\right] d\bar{z}$$

$$= \int q(\bar{z}|\theta) \log(\bar{p}(\bar{z}|\theta)) d\bar{z} - \mathbb{D}_{KL}[q(\bar{z}|\theta) \| p(\bar{z})] \quad (17.18)$$

$$\mathbb{D}_{KL}[q(\bar{z}|\theta) \| p(\bar{z})] = \int q(\bar{z}|\theta) \log\left[\frac{q(\bar{z}|\theta)}{p(\bar{z})}\right] d\bar{z}$$

$$\stackrel{17.15}{=} \int q(\bar{z}|\theta) \log\left[q(\bar{z}|\theta) \left(\frac{p(\bar{x}|\theta, \bar{z})}{p(\bar{x}|\theta)}\right)^{-1}\right] d\bar{z}$$

$$= \int q(\bar{z}|\theta) \log\left[q(\bar{z}|\theta) (p(\bar{x}|\theta, \bar{z}) p(\bar{z}))^{-1} p(\bar{x}|\theta)\right] d\bar{z}$$

$$\quad \quad \quad \int q(\bar{z}|\theta) d\bar{z} = 1$$

$$= \underbrace{\int q(\bar{z}|\theta) \log\left[\frac{q(\bar{z}|\theta)}{p(\bar{z})}\right] d\bar{z}}_{\mathbb{D}_{KL}[q(\bar{z}|\theta) \| p(\bar{z})]} + \underbrace{\int q(\bar{z}|\theta) \log[p(\bar{x}|\theta)] d\bar{z}}_{\mathbb{E}_{q(\bar{z}|\theta)}[\log[p(\bar{x}|\theta)]]}$$

$$- \underbrace{\int q(\bar{z}|\theta) \log[p(\bar{x}|\theta, \bar{z})] d\bar{z}}_{\mathbb{E}_{q(\bar{z}|\theta)}[\log[p(\bar{x}|\theta, \bar{z})]]}$$

$$\Rightarrow \mathbb{E}_{q(\bar{z}|\theta)}[\log[p(\bar{x}|\theta, \bar{z})]] - \mathbb{D}_{KL}[q(\bar{z}|\theta) \| p(\bar{z})] \leq \log[p(\bar{x}|\theta)]$$

$$= \log[p(\bar{x}|\theta)] - \mathbb{D}_{KL}[q(\bar{z}|\theta) \| p(\bar{z})] \quad \square$$

17.5

$$\frac{\partial}{\partial \phi} \mathbb{E}_{p(\bar{x}|\phi)}[f(\bar{x})] = \frac{\partial}{\partial \phi} \int f(\bar{x}) p(\bar{x}|\phi) d\bar{x} \quad (\text{x discrete})$$

$$= \sum_{\bar{x}} f(\bar{x}) \frac{\partial}{\partial \phi} p(\bar{x}|\phi)$$

$$= p(\bar{x}|\phi) \frac{\partial}{\partial \phi} \log[p(\bar{x}|\phi)]$$

$$= \sum_{\bar{x}} f(\bar{x}) \frac{\partial}{\partial \phi} \log[p(\bar{x}|\phi)] p(\bar{x}|\phi)$$

$$= \mathbb{E}_{p(\bar{x}|\phi)}\left[f(\bar{x}) \frac{\partial}{\partial \phi} \log[p(\bar{x}|\phi)]\right]$$

Using Monte Carlo approximation: $\mathbb{E}_{p(\bar{x}|\phi)}[g(\bar{x})] \approx \frac{1}{I} \sum_{i=1}^I g(\bar{x}_i)$

with $g := f(\bar{x}) \frac{\partial}{\partial \phi} \log[p(\bar{x}|\phi)]$

we arrive immediately at: $\frac{\partial}{\partial \phi} \mathbb{E}_{p(\bar{x}|\phi)}[f(\bar{x})]$

$$\approx \frac{1}{I} \sum_{i=1}^I f(\bar{x}_i) \frac{\partial}{\partial \phi} \log[p(\bar{x}_i|\phi)] \quad \square$$

17.6 It is better to use spherical linear interpolation rather than regular linear interpolation when moving between points in the latent space because of the curse of dimensionality. (everything is far apart in higher dimensions!)

$$17.7 \quad p(\bar{x}|\theta) = \sum_{k=1}^K \lambda_k \text{Norm}_x[\mu_k, \Sigma_k]$$

$$p(\bar{x}|\theta, \bar{z}) = \text{Norm}_x[\mu_{\bar{z}}, \Sigma_{\bar{z}}]$$

$$\text{and } p(\bar{z}|\theta) = \text{cat}_{\bar{z}}[\bar{\lambda}]$$

$$\text{so } p(\bar{x}|\theta) = \sum_{k=1}^K p(\bar{x}, \bar{z}=k|\theta)$$

$$= \sum_{k=1}^K p(\bar{x}|\bar{z}=k, \theta) \cdot p(\bar{z}=k|\theta)$$

$$= \sum_{k=1}^K \lambda_k \text{Norm}_x[\mu_k, \Sigma_k]$$

$$\text{Update rules: } \lambda_k^{(t+1)} = \frac{\bar{\lambda}_{k-1}^T \bar{r}_k}{\sum_{j=1}^K \bar{\lambda}_{j-1}^T \bar{r}_j}$$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^I \bar{r}_{ik} x_i}{\sum_{i=1}^I \bar{r}_{ik}}$$

$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^I \bar{r}_{ik} (x_i - \mu_k^{(t+1)}) (x_i - \mu_k^{(t+1)})^T}{\sum_{i=1}^I \bar{r}_{ik}}$$